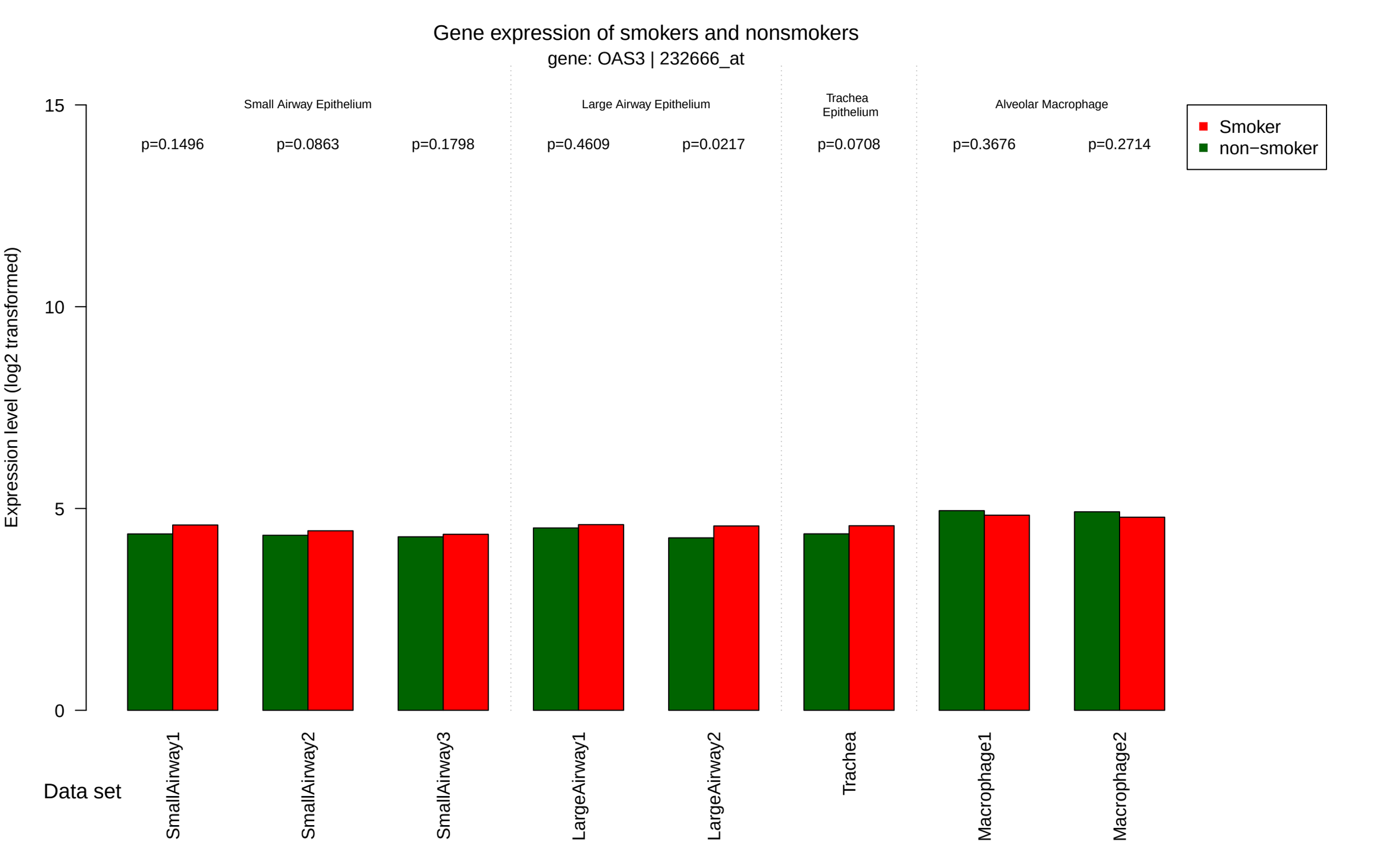
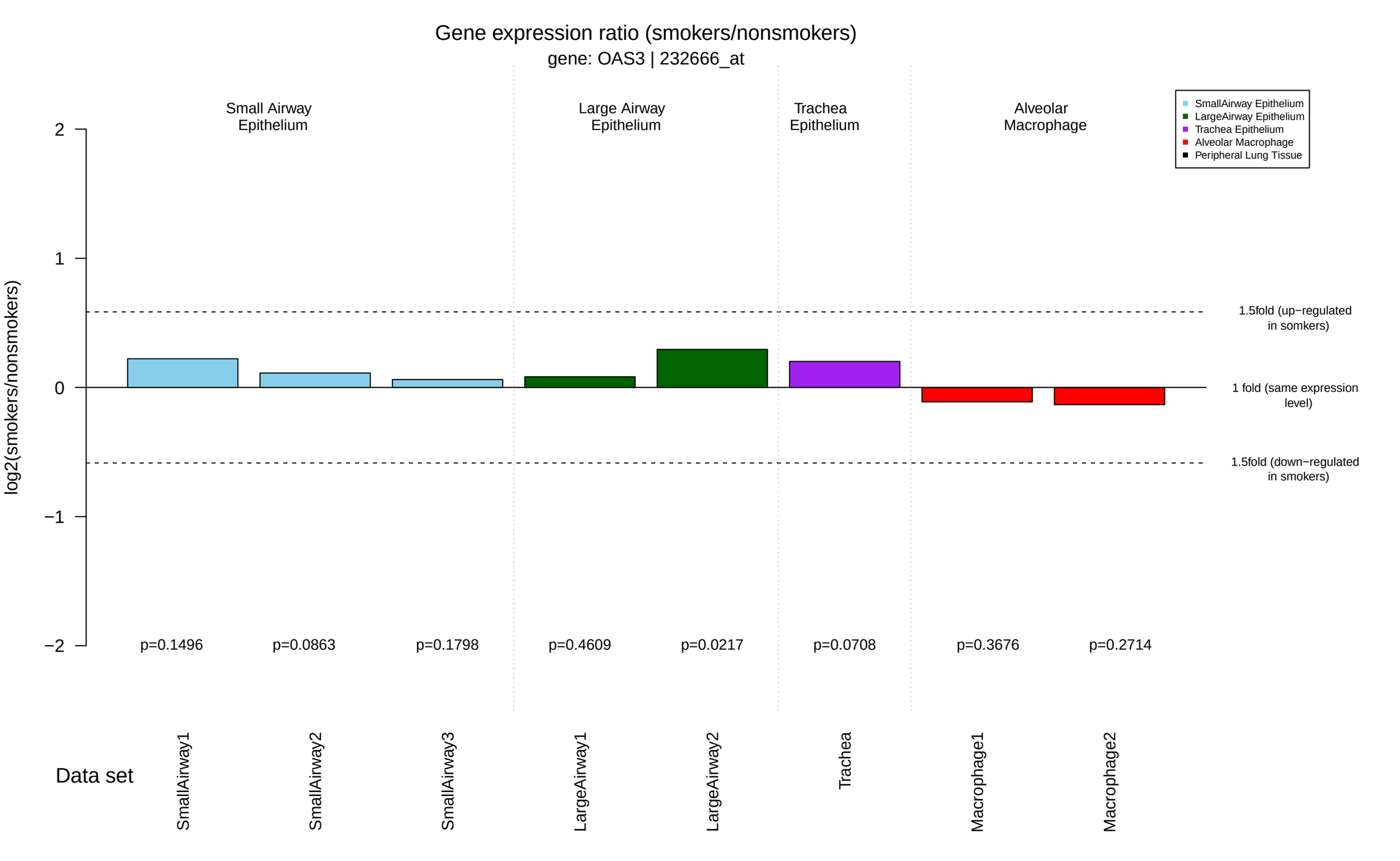
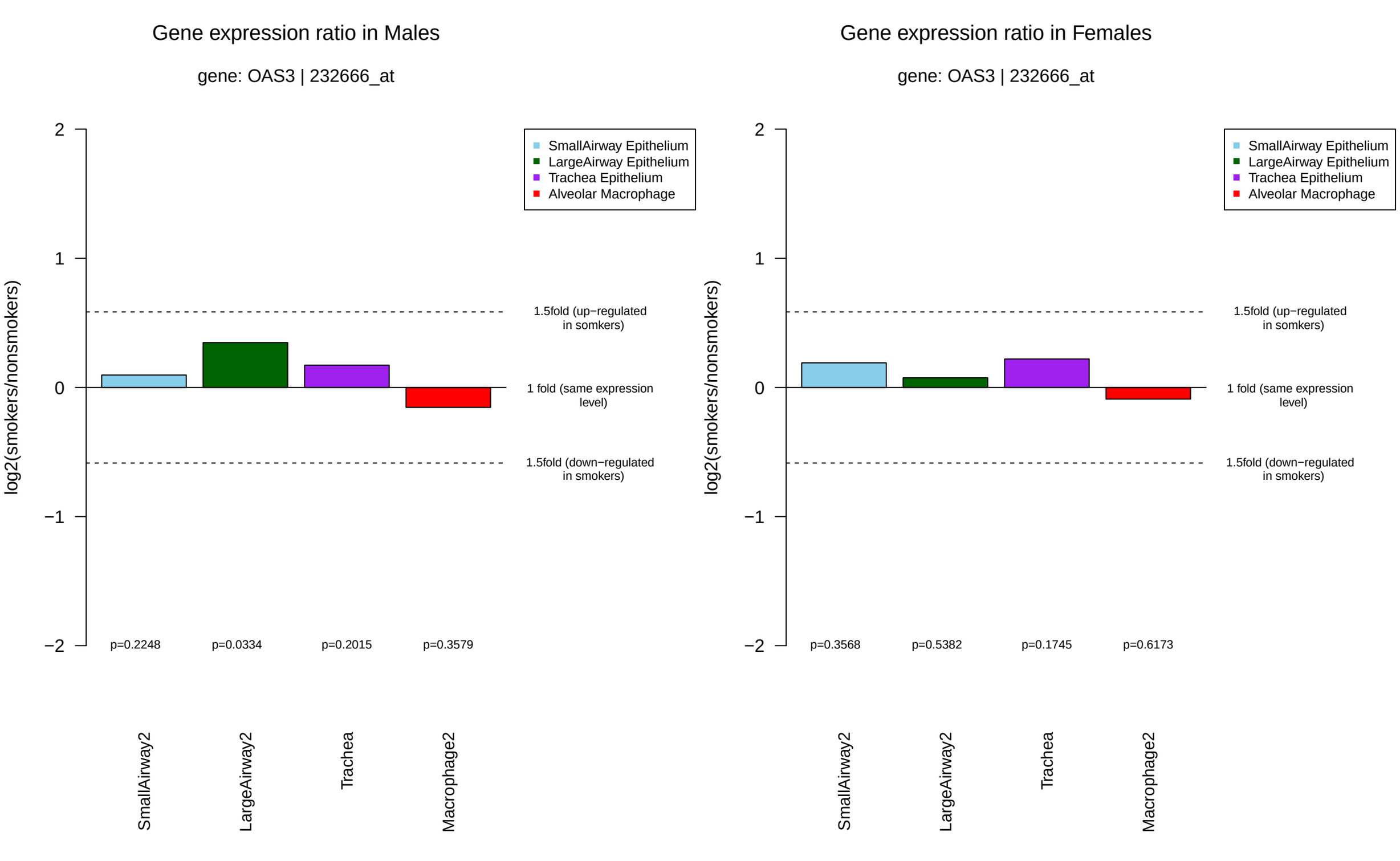
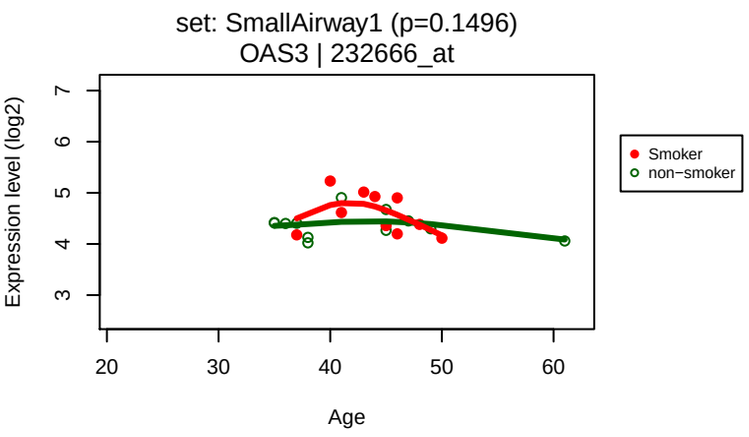
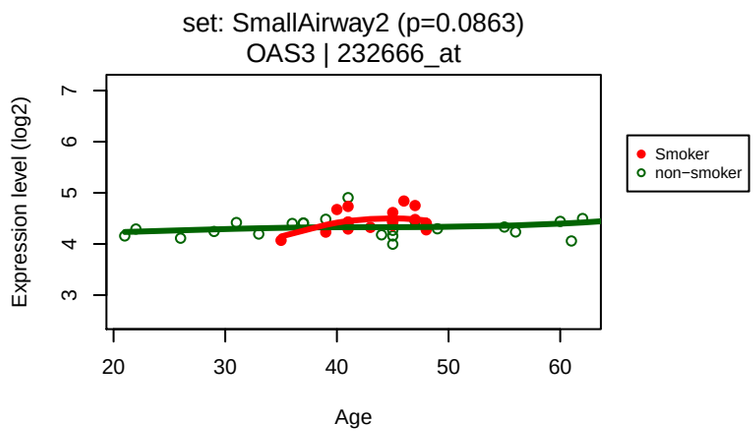
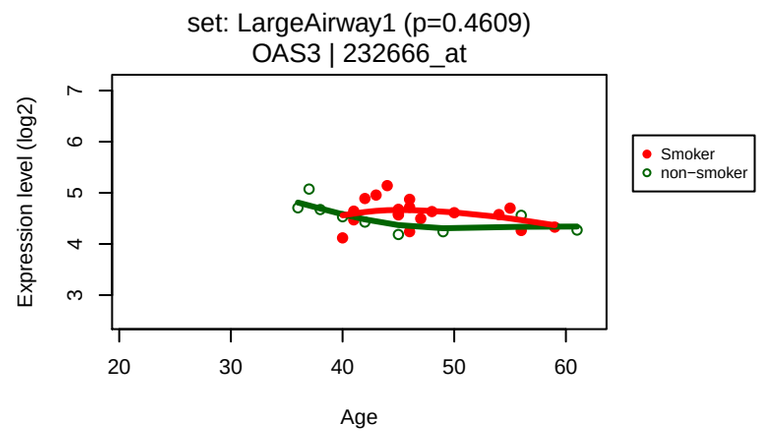

Gene expression of smokers and nonsmokers
gene: OAS3 | 232666_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1496
p=0.0863
p=0.1798
p=0.4609
p=0.0217
p=0.0708
p=0.3676
p=0.2714
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

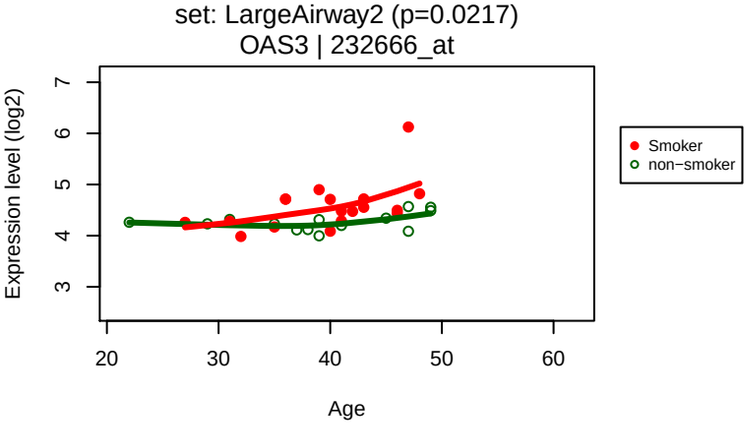

Gene expression ratio (smokers/nonsmokers)
gene: OAS3 | 232666_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1496
p=0.0863
p=0.1798
p=0.4609
p=0.0217
p=0.0708
p=0.3676
p=0.2714
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

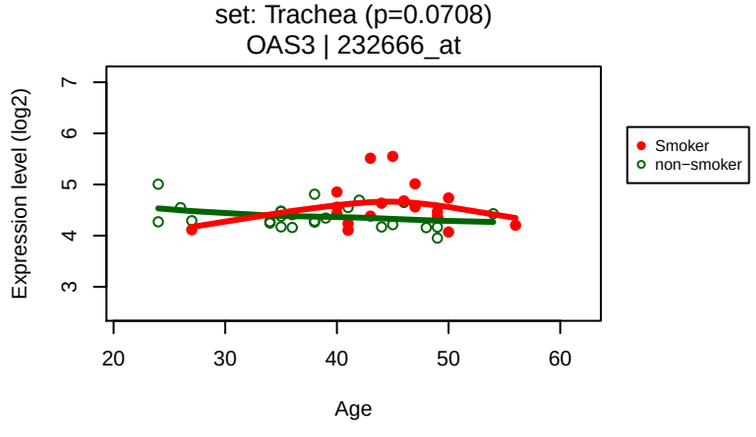

Gene expression ratio in Males
gene: OAS3 | 232666_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2248
p=0.0334
p=0.2015
p=0.3579
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OAS3 | 232666_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3568
p=0.5382
p=0.1745
p=0.6173
SmallAirway2
LargeAirway2
Trachea
Macrophage2

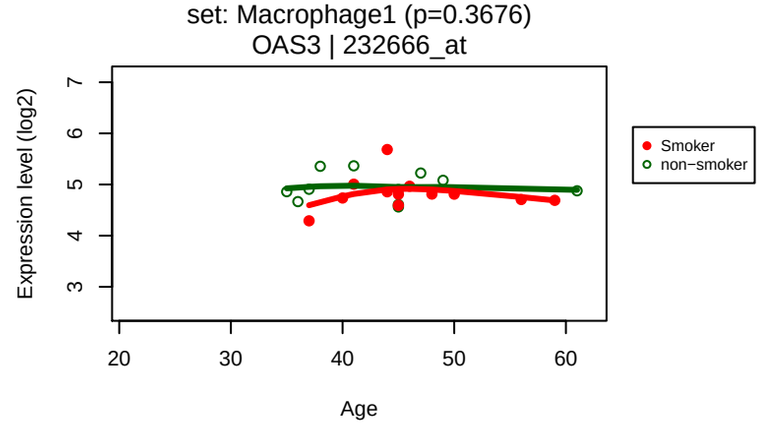

set: SmallAirway1 (p=0.1496)
OAS3 | 232666_at
Expression level (log2)
Age
Smoker
non−smoker

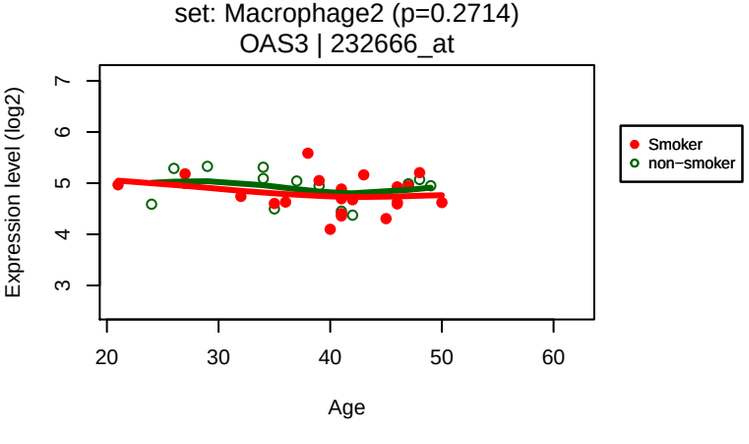

set: SmallAirway2 (p=0.0863)
OAS3 | 232666_at
Expression level (log2)
Age
Smoker
non−smoker

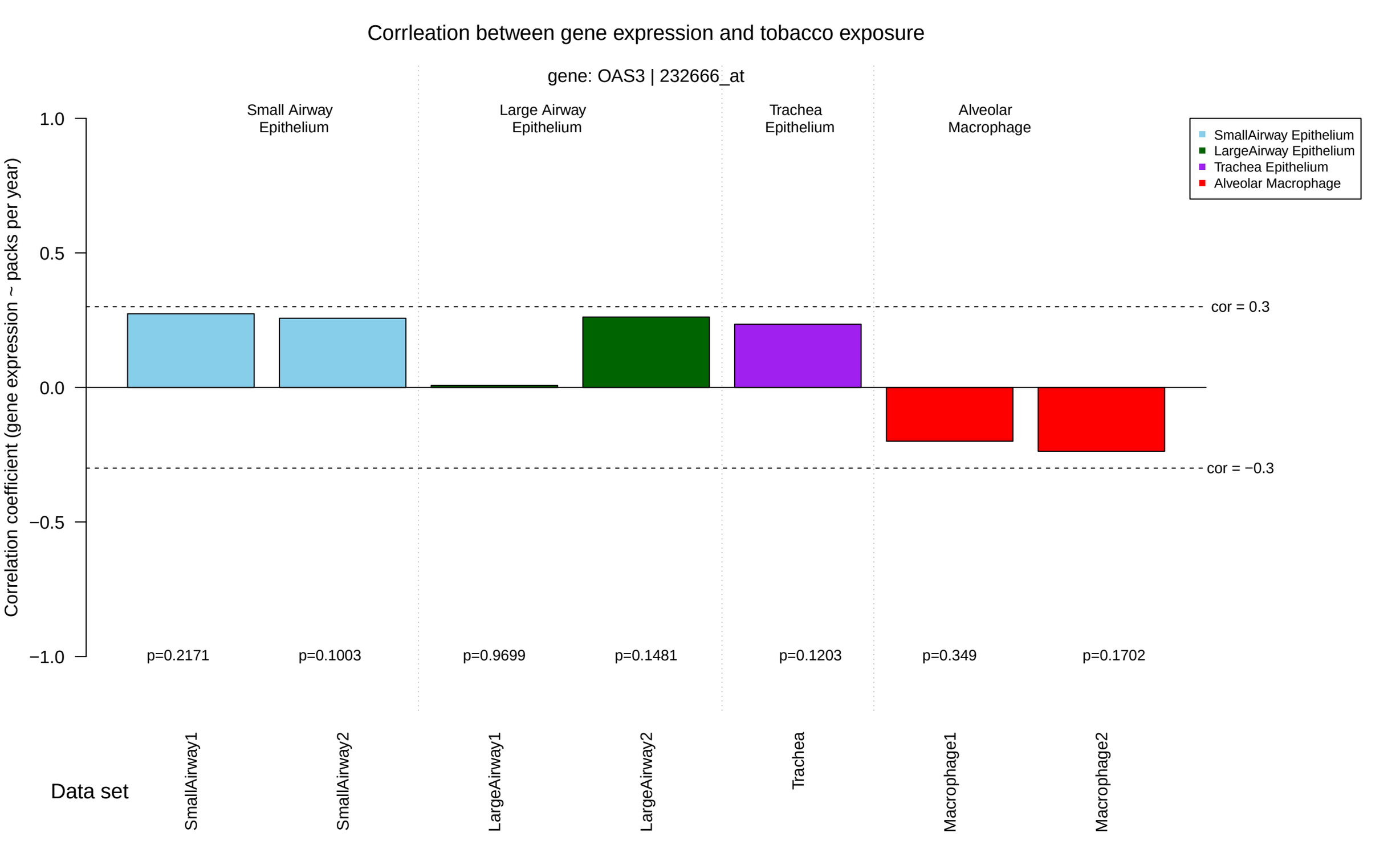

set: LargeAirway1 (p=0.4609)
OAS3 | 232666_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0217)
OAS3 | 232666_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0708)
OAS3 | 232666_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3676)
OAS3 | 232666_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2714)
OAS3 | 232666_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OAS3 | 232666_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2171
p=0.1003
p=0.9699
p=0.1481
p=0.1203
p=0.349
p=0.1702
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2